

1a Formal Notation (from chapter 1.1.5)

Def Sequence (tuple): $(x_j)_{j=1}^n = (x_1, x_2, \dots, x_n)$

Def Range: $[n] = \{1, 2, \dots, n\}$

Def Sum: accumulation over index $i = m, m + 1, \dots, n$.

$$\sum_{i=m}^n x_i = x_m + x_{m+1} + \dots + x_n$$

Empty sum equals 0 (can be vector or matrix)

- **Number of Terms:** $n - m + 1$
- Same with product $\prod_{i=m}^n x_i = x_1 x_2 \dots x_n$.

Def Sets: unordered collection; ignores duplicates; notation: $\{\dots\}$

Def Natural numbers: $\mathbb{N} = \{0, 1, 2, \dots\}$

1b Vector Basics

Vector Definition

Def 1.1 Let $m \in \mathbb{N}$. An **m-dimensional coordinate vector** is element of $\mathbb{R}^m$.

• Written as **column vector:** $v = \begin{pmatrix} v_1 \\ \vdots \\ v_m \end{pmatrix}$.

- **Zero vector** (0): All coordinates 0.
- **Zero-dimension:** $\mathbb{R}^0 = \{\{\}\}$ only contains empty tuple.

Vector Addition

Def 1.2 Let $v, w \in \mathbb{R}^m$. The sum is: $v + w := \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_m + w_m \end{pmatrix} \in \mathbb{R}^m$

- **Geometric View:** Combines movement of v and w .
- **Associative:** $(u + v) + w = u + (v + w)$.
- **Commutative:** $v + w = w + v$.

Scalar Multiplication

Def 1.3 Let $v \in \mathbb{R}^m$ and $\lambda \in \mathbb{R}$. The scalar multiple is $\lambda v := \begin{pmatrix} \lambda v_1 \\ \vdots \\ \lambda v_m \end{pmatrix}$.

- **Geometric View:** Scales length by factor $|\lambda|$ (neg. **flips** direction).

Scalar Product

Def 1.9 Let $v, w \in \mathbb{R}^m$. The scalar product is: $v \cdot w := \sum_{i=1}^m v_i w_i$

Obs 1.10 Let $u, v, w \in \mathbb{R}^m, \lambda \in \mathbb{R}$.

1. **Symmetry:** $v \cdot w = w \cdot v$
2. **Linearity:** $(\lambda v) \cdot w = \lambda(v \cdot w) = v \cdot (\lambda w)$
3. **Distributivity:** $u \cdot (v + w) = u \cdot v + u \cdot w$
4. **Positive-definiteness:** $v \cdot v \geq 0$, equality iff $v = 0$.

Transpose

Def 1.20 The **covector** (transpose of v) $v^T \in (\mathbb{R}^m)^*$ is the function $v^T : \mathbb{R}^m \rightarrow \mathbb{R}, x \mapsto \sum_{i=1}^m v_i x_i$.
Written as **row vector:** $v^T = (v_1 \dots v_m)$ and reversable $(v^T)^T := v$.

Def 1.19 Covector-vector multiplication: $v^T w = v \cdot w$

Length (Euclidean Norm)

Def 1.11 For $v \in \mathbb{R}^m$, the **Euclidean norm** is: $\|v\| := \sqrt{v \cdot v}$

- Generalizes **Pythagorean theorem** (measures length).
- **Unit vector:** $\|u\| = 1$. In $\mathbb{R}^2$, lies on **unit circle**.
- **Normalize:** Create unit vector via $u = \frac{v}{\|v\|}$ (for $v \neq 0$).
- **Std. unit vectors** (e_i): 1 at element i , 0 elsewhere. Points along coordinate axes.

Angles & Orthogonality

Def 1.14 For nonzero $v, w \in \mathbb{R}^m$, the **angle** is unique $\alpha \in [0, \pi]$ where

$$\cos(\alpha) = \frac{v \cdot w}{\|v\| \|w\|} \in [-1, 1]$$

- Valid range $[-1, 1]$ guaranteed by **Cauchy-Schwarz**.

Def 1.15 For $v, w \in \mathbb{R}^m$ are **orthogonal** if $v \cdot w = 0$.

- **Orthogonal:** Nonzero vectors Angle is 90° ($\cos = 0$).
- **Zero vector** is orthogonal to everything.

Def 1.16 Let $d \neq 0$. A **hyperplane** through origin contains all vectors orthogonal to d : $H_d = \{v \in \mathbb{R}^m : v \cdot d = 0\}$

Fundamental Inequalities

Lem 1.12 (Cauchy-Schwarz) Let $v, w \in \mathbb{R}^m$ then $|v \cdot w| \leq \|v\| \|w\|$.
Equality iff one vector is scalar multiple of the other.

Lem 1.17 (Triangle ineq.) Let $v, w \in \mathbb{R}^m$ then $\|v + w\| \leq \|v\| + \|w\|$.

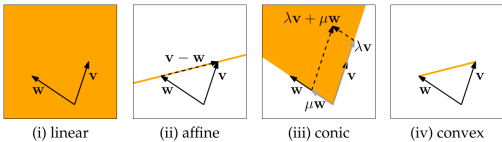
1c Vector Linearity

Linear Combinations

Def 1.4 If $v_1, \dots, v_n \in \mathbb{R}^m$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, then $\sum_{i=1}^n \lambda_i v_i$ is a **linear combination**.

Def 1.7 A linear combination $\sum \lambda_i v_i$ is:

1. **Affine:** if $\sum \lambda_i = 1$.
2. **Conic:** if $\forall j, \lambda_j \geq 0$.
3. **Convex:** if both affine and conic ($\sum \lambda_i = 1$ and $\lambda_j \geq 0$).



Spanning R^2 Prove

- **Strategy:** Given specific $v, w \in \mathbb{R}^2$ solve for λ, μ in $u = \lambda v + \mu w$ where $u \in \mathbb{R}^2$ is arbitrary.
- **Row Picture:** Equations are lines in $\lambda\mu$ -plane. Solution is **intersection**. Non-parallel lines $\rightarrow$ unique intersection.
- **Column Picture:** v, w form **skewed coordinate system**. Target u is diagonal of **parallelogram** formed by $\lambda v, \mu w$.

Linear (In)dependence

Def 1.21 Vectors $v_1, \dots, v_n$ are **linearly dependent** if at least one is a linear combination of others: $\exists k \in [n], \lambda_j$ such that $v_k = \sum_{j \neq k} \lambda_j v_j$. Otherwise, they are **linearly independent**.

Lem 1.22 / 1.23 Let $v_1, \dots, v_n \in \mathbb{R}^m$.

- **Dependent** $\Leftrightarrow$ **Non-trivial** comb. of zero exists (some $\lambda_j \neq 0$ where $\sum \lambda_j v_j = 0$) $\Leftrightarrow$ One vector is lin. comb. of **previous** ones.
- **Independent** $\Leftrightarrow$ Only **trivial** comb. of zero exists ($\sum \lambda_j v_j = 0 \Rightarrow$ all $\lambda_j = 0$) $\Leftrightarrow$ None is lin. comb. of **previous** ones.

Lem 1.24 If $v_1, \dots, v_n \in \mathbb{R}^m$ are **linearly independent**, any linear combination $v \in \mathbb{R}^m$ has **unique** coefficients.

- **Dependent:** has **zero vector** (0) or length $n > m$ (in $\mathbb{R}^m$).
- **Independent:** **Empty** sequence.

Span of Vectors

Def 1.25 The **span** is the set of all possible linear combinations:

$$\text{Span}(v_1, \dots, v_n) := \left\{ \sum_{j=1}^n \lambda_j v_j : \lambda_j \in \mathbb{R} \right\}$$

- 1 **non-zero** vector: **Line** through origin.
- 2 **non-collinear** vectors: **Plane** through origin.
- Empty sequence: **Point** $\{0\}$.

Lem 1.26 / 1.27 Adding a lin. comb. of existing vectors or **removing** a lin. comb. of others **does not change** the Span.

Lem 1.28 m **linearly independent** vectors in $\mathbb{R}^m$ span the entire space: $\text{Span}(v_1, \dots, v_m) = \mathbb{R}^m$

2a Matrix Basics

Matrix Definition

Def 2.1 An $m \times n$ **matrix** $A \in \mathbb{R}^{m \times n}$ has m rows and n columns.

- **Entry-wise:** $A \Rightarrow a_{ij}$ (row i , column j).
- **Dot-free:** $A = [a_{ij}]_{i=1, j=1}^{m, n}$.
- **Column Notation:** $A = (v_1 \dots v_n)$ with $v_i \in \mathbb{R}^m$.
- **Row Notation:** $A = \begin{pmatrix} u_1^T \\ \vdots \\ u_m^T \end{pmatrix}$ with $u_i^T \in (\mathbb{R}^n)^*$.

Def 2.2 Let $A, B \in \mathbb{R}^{m \times n}, \lambda \in \mathbb{R}$.

- **Addition:** $A + B := [a_{ij} + b_{ij}]_{i=1, j=1}^{m, n}$ (element-wise).
- **Scalar Mult:** $\lambda A := [\lambda a_{ij}]_{i=1, j=1}^{m, n}$ (element-wise).
- **Zero Matrix:** 0 contains all zeros.

Def 2.12 The **transpose** of $A \in \mathbb{R}^{m \times n}$ is $A^T \in \mathbb{R}^{n \times m}$ where $a_{ij}^T = a_{ji}$.

Obs 2.13 Double Transpose: $(A^T)^T = A$

Special Matrix Types

Def 2.3 For $A \in \mathbb{R}^{m \times m}$:

1. **Identity (I):** Diagonal entries 1, off-diagonal 0. Defined by **Kronecker delta** δ_{ij} (1 if $i = j$, else 0).
2. **Diagonal:** Off-diagonal entries are 0 ($a_{ij} = 0$ for $i \neq j$).
3. **Upper triangular:** Entries below diagonal are 0 ($a_{ij} = 0$ for $i > j$).
4. **Lower triangular:** Entries above diagonal are 0 ($a_{ij} = 0$ for $i < j$).
5. **Symmetric:** $a_{ij} = a_{ji}$ for all i, j .

Subspaces

Def 2.9 The **column space** $C(A)$ is the span of the columns:

$$C(A) := \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$$

Def 2.14 The **row space** is the transpose: $R(A) := C(A^T) \subseteq \mathbb{R}^n$

Def 2.17 The **nullspace** is the set of inputs mapped to zero:

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n$$

- **Independence:** $N(A) = \{0\} \Leftrightarrow$ columns of A are lin. independent.

Rank

Def 2.10 Let $A = (v_1 \dots v_n)$.

- **Independent Column:** v_j is **not** a linear combination of **preceding** columns $(v_1, \dots, v_{j-1})$. Same for **rows** via transpose.
- **Rank:** $\text{rank}(A)$ is the number of independent columns.

- **Full Rank:** $\text{rank}(A) = n \Leftrightarrow$ all columns independent.
- **Zero Rank:** $\text{rank}(A) = 0 \Leftrightarrow A$ is zero matrix.
- **Rank Matches:** Column rank = Row rank.

Lem 2.11 The **indep. col.** span the column space: $C(A) = C(C_{\text{indep}})$

Lem 2.15 $\text{rank}(A) = 1 \Leftrightarrow$ Non-zero A is outer product of two vectors $v \in \mathbb{R}^m, w \in \mathbb{R}^n$: $A = vw^T$ (entries $v_i w_j$)

2b Matrix Transformations

Matrix-Vector Multiplication

Def 2.4 Let $A = (v_1 \dots v_n)$ and $x \in \mathbb{R}^T$: $Ax := \sum_{j=1}^n x_j v_j \in \mathbb{R}^m$
 Ax is the **linear combination** of A 's columns.

Obs 2.8 **Row Notation:** Let A have rows $u_1^T, \dots, u_m^T$: $Ax = \begin{pmatrix} u_1^T x \\ \vdots \\ u_m^T x \end{pmatrix}$

The i -th entry is the **scalar product** of the i -th row and x .

Obs 2.6 **Table Notation:** The entries are $(Ax)_i = \sum_{j=1}^n a_{ij} x_j$.

Obs 2.5 **Implications:**

1. b is a **linear comb.** of columns of $A \Leftrightarrow Ax = b$ has a solution.
2. A is **full rank** $\Leftrightarrow Ax = 0$ has only the trivial solution $x = 0$.

Cor 2.7 **Identity:** $Ix = x$ for all $x \in \mathbb{R}^m$.

Matrices as Linear Transformations

Def 2.18 The **matrix transf.** is the function $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Lem 2.19 **Linearity:** $A(\lambda_1 x_1 + \lambda_2 x_2) = \lambda_1 Ax_1 + \lambda_2 Ax_2$.

Obs 2.22 Every matrix transformation is a linear transformation.

Thr 2.26 Every linear transformation corresponds to a **unique matrix** $A = (T(e_1) \ T(e_2) \ \dots \ T(e_n))$

General Linear Transformations

Def 2.21 Func. T is **linear** if: $T(\lambda_1 x_1 + \lambda_2 x_2) = \lambda_1 T(x_1) + \lambda_2 T(x_2)$
(**Functional** if maps to $\mathbb{R}$, **Transformation** if maps to $\mathbb{R}^m$).

Lem 2.23 T is linear $\Leftrightarrow$ for all vectors x, x' and scalars λ :

1. **Additivity:** $T(x + x') = T(x) + T(x')$
2. **Homogeneity:** $T(\lambda x) = \lambda T(x)$

Lem 2.24 / 2.25 T linear $\Rightarrow T(0) = 0$ and $T(\sum \lambda_j x_j) = \sum \lambda_j T(x_j)$

Kernel and Image

Generalizations of Nullspace and Column Space to linear transforms.

Def 2.27 **Image:** $\text{Im}(T) := \{T(x) : x \in \mathbb{R}^n\}$ (All possible outputs).

Kernel: $\text{Ker}(T) := \{x \in \mathbb{R}^n : T(x) = 0\}$ (Inputs mapped to zero).

Obs 2.28 / 2.29 **Relation:** $\text{Im}(T) = C(A)$ and $\text{Ker}(T) = N(A)$.

2c Matrix Multiplication

Def 2.33 **Function Composition:** $f \circ g : X \rightarrow Z, x \mapsto f(g(x))$.

Matrix Multiplication Concepts

For $A \in \mathbb{R}^{a \times n}$ and $B \in \mathbb{R}^{n \times b}$

Lem 2.34 / 2.35 / 2.37 **Matrix Composition:** For linear T_A and T_B :

1. The composition $T_A \circ T_B$ is a matrix transformation.
2. It corresponds to the **unique** matrix product: $T_A \circ T_B = T_{AB}$.

Lem 2.36 **Column Notation:** $AB := [Ab_1, \dots, Ab_b]$ (B 's columns)

Obs 2.45 **Row Notation:** $AB = \begin{pmatrix} a_1^T B \\ \vdots \\ a_a^T B \end{pmatrix}$ (A 's rows)

Obs 2.38 / 2.39 **Entry-wise:** $(AB)_{ij} = a_i \cdot b_j = \sum_{l=1}^n a_{il} b_{lj}$

Lem 2.40 / 2.41 / 2.42 Properties:

- **Neutrality:** $IA = A$ and $AI = A$.
- **Distrib.:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
- **Associativity:** $(AB)C = A(BC)$.
- **Transpose:** $(AB)^T = B^T A^T$. (reverses order)

Mixed Multiplications (“The Zoo”)

- **Unifying View:** Vectors are $m \times 1$, Covectors are $1 \times n$ matrices.

Def 2.43 Covector-Matrix: $y^T A = (y^T v_1 \quad \dots \quad y^T v_n) \in (\mathbb{R}^n)^*$

Def 2.44 Outer Product: For $v \in \mathbb{R}^m, w \in \mathbb{R}^n$: $vw^T = [v_i w_j]_{i,j}$

CR-Decomposition

- **Thr 2.46** Let $A \in \mathbb{R}^{m \times n}$ have **rank** r . Unique decomp. $A = CR'$:
 - $C \in \mathbb{R}^{m \times r}$. Contains the r **independent columns** of A .
 - $R' \in \mathbb{R}^{r \times n}$: Unique coefficients to express A 's columns via C .

- **Compression** (only low rank): store $(m + n)r$ entries instead of mn .

2d Matrix Invertibility

Def 2.48 Let $f : X \rightarrow Y$ be a function.

- **Injective:** Every $y \in Y$ has **at most one** pre-image $x \in X$.
- **Surjective:** Every $y \in Y$ has **at least one** pre-image $x \in X$.
- **Bijective:** Both injective and surjective (exactly one pre-image).
- **Inverse:** If bijective, $f^{-1} : Y \rightarrow X$ maps $y \mapsto x$ where $f(x) = y$.

- **Lem 2.49** If f is bijective $\Rightarrow f^{-1}$ is also bijective and $(f^{-1})^{-1} = f$
- **Cancellation:** $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

Invertibility and Inverse Matrices

- **Bijective** transformation $\Leftrightarrow$ **Invertible** matrix $\Rightarrow$ matrix **square**.

Lem 2.50 Wide Mat.: If $A \in \mathbb{R}^{m \times n}$ with $m < n$, T_A is **not injective**.

Lem 2.51 Tall Mat.: If $A \in \mathbb{R}^{m \times n}$ with $m > n$, T_A is **not surjective**.

Lem 2.52 If linear transf. T is bijective, T^{-1} is also a **linear transf.**.

- **Def 2.53 / 2.55 / 2.57** For $A \in \mathbb{R}^{m \times m}$: A is **invertible** $\Leftrightarrow \exists B$ s.t. $BA = I \Leftrightarrow T_A$ is **bijective** $\Leftrightarrow$ Columns of A are **lin. indep.**.
- **Inverse:** The **unique** $B = A^{-1}$.
- **Singular:** A is not invertible.

Properties and Formulas

Lem 2.54 / 2.56 Commutativity: $A^{-1}A = AA^{-1} = I$

Obs 2.58 If A is invertible, A^{-1} is invertible and $(A^{-1})^{-1} = A$.

Lem 2.59 If A, B invertible, AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$..

Lem 2.60 If A invertible, A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

- $[a]^{-1} = \begin{bmatrix} 1 & \\ & a \end{bmatrix}$ iff $a \neq 0$.
- $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ iff $\det(A) \neq 0$.

3a Solving Linear Equations Ax = b

Def 3.1 System of Linear Equations: m sum equations, n variables.

- **Matrix-Vector Form:** $Ax = b$ where $A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^n, b \in \mathbb{R}^m$
- **Row Picture:** Rows u_i^T of A define **hyperplanes** $u_i^T x = b_i$.
 - Solution is the **intersection point** of all m hyperplanes.
- **Column Pic.**: Find coeff. x_j for **linear combination** $b = \sum x_j a_j$
 - Solution exists $\Leftrightarrow b \in C(A)$ (Column Space).

3b Gauss Elimination

Transform **square** $Ax = b$ into equivalent **upper triangular** $Ux = c$.

1. Elimination Procedure

- For $j \in [m - 1]$ use pivot u_{jj} to **eliminate entries below**.
 - **Zero Pivot:** Swap with lower row having non-zero entry.
 - **Failure:** Pivot is 0 and all below are $0 \Rightarrow A$ is **singular**.
2. **Back Substitution:** Solve $Ux = c$ from bottom up ($x_m \rightarrow x_1$).
- **Requirement:** Pivots must be **non-zero**.

Row Operations & Properties

Row ops. O_i like left-multip. by **invertible matrix** $M = O_1 \dots O_q$.

- **Lem 3.2 - 3.6 Invariant Properties** (under invertible M):
 - Systems $Ax = b$ and $MAx = Mb$ have the **same solutions**.
 - **Nullspace:** $N(A) = N(MA)$.
 - **Independence:** A cols. are indep. $\Leftrightarrow MA$ cols. are indep.
 - **Row Space:** $R(A) = R(MA)$.
 - **Rank:** A and MA have same rank.

- **NOT Invariant:** Column Space $(C(A) \neq C(MA)$ usually).

Success, Failure & Runtime

Thr 3.7 For square A , equivalent statements: Gauss el. **succeeds** $\Leftrightarrow A$ columns are **linearly independent** $\Leftrightarrow A$ is **invertible**.

Thr 3.8 Cols **indep.** $\Rightarrow$ Unique solution; Cols **dependent** $\Rightarrow$ Fails.

Thr 3.9 - 3.12 Runtime (m variables):

- **Elimination:** $O(m^3)$; **Back Substitution:** $O(m^2)$.
- **Inverse / Multiple RHS:** $O(m^3)$.

- **Faster:** Strassen $O(m^{2.81})$ exists but rarely used (big constant).

3 Gauss-Jordan Elimination

- Transform A into unique **Reduced Row Echelon Form** (RREF).
- **Operations:** Subtraction, Exchange, **Division** (create pivot 1).

Def 3.13 R is in RREF if exist col. indices $1 \leq j_1 < \dots < j_r \leq n$ s.t.:

1. $\forall i \in [r]$ column j_i is **standard unit vector** e_i
2. Entries $(R)_{ij}$ “below staircase” are 0 ($i > r$ OR $i \leq r, j < j_i$).

- **Notation:** RREF($j_1, \dots, j_r$) (matrix with r pivots).
- **Identity** I_m is RREF($1, \dots, m$); **Zero matrix** is RREF($\emptyset$).

Lem 3.14 RREF($j_1, \dots, j_r$) has **indep. col.** $j_1, \dots, j_r$ and **rank** r .

The Gauss-Jordan Algorithm

- **Process:** Transform augmented $(A|b)$ to $(R|c)$ column by column.
 1. **Normalize Pivot:** Divide row by pivot value (pivot becomes 1).
 2. **Eliminate:** Row subtraction to zero **above** and **below** pivot.
 3. **Skip:** If column cannot form pivot (all 0 below), skip it.
- **Direct Solution** ($Rx = c$): Set $x_{j_i} = c_i$ (pivot vars), set others to 0.
- **No solution** if $c_i \neq 0$ for any zero-row i in R .

Thr 3.16 Elimination Runtime: For m systems $Ax = b_j$ (n vars): Gauss-Jordan returns equivalent $Rx = c_j$ in $O(m^2(m + n))$.

Thr 3.15 Direct Sol. Runtime: Solving $Rx = c$ takes $O(m + n)$.

Theoretical Results & Applications

Thr 3.17 $(A|I)$ to $(R|M) \Rightarrow M$ is **invertible**, $R = MA$, R is RREF

- **Thr 3.18 CR-Decomposition:** Every A has a **unique** RREF R .
- Indices $j_1 \dots j_r$ are the **independent columns** of A (form C).
- R contains R' (top r rows) from **CR decomposition** $A = CR'$.

Thr 3.19 $(A|I)$ to $(R|M)$: A is **invertible** $\Leftrightarrow R = I$ (then $M = A^{-1}$)

Thr 3.20 Solving $Ax = b$ Strategy: Efficient for multiple b 's:

1. **Preprocess:** Run on $(A|I)$ to get $(R|M)$. Cost: $O(m^2(m + n))$.
2. **Solve:** Compute $c = Mb$, then solve $Rx = c$. Cost: $O(m^2 + n)$.

4. The Abstract Vector Space

Def 4.1 $(V, +, \cdot)$ is a **Vector Space** which $\forall u, v, w \in V$ and $\lambda, \mu \in \mathbb{R}$:

1. **Commutativity:** $v + w = w + v$
2. **Associativity:** $u + (v + w) = (u + v) + w$
3. **Zero vector:** Exists 0 s.t. $v + 0 = v$
4. **Negative vector:** Exists $-v$ s.t. $v + (-v) = 0$
5. **Identity:** $1 \cdot v = v$
6. **Compatibility of multiplications:** $(\lambda\mu)v = \lambda(\mu v)$
7. **Distributivity (+):** $\lambda(v + w) = \lambda v + \lambda w$
8. **Distributivity ($\mathbb{R}$):** $(\lambda + \mu)v = \lambda v + \mu v$

Lem 4.6 / 4.7 / 4.10 Basic Properties:

- **Unique Zero:** V contains exactly one zero vector.
- **Unique Negative:** Every v has exactly one inverse $-v$.
- **Zero Scaling:** $0 \cdot v = 0$ (provable, not axiom).

Obs 4.2 / 4.5 $\mathbb{R}^m$ (tuples) and $\mathbb{R}^{m \times n}$ (matrices) are vector spaces.

Def 4.3 / 4.4 Polynomials $\mathbb{R}[x]$ is a vector space.

- **Polynomial:** $p = \sum_{i=0}^m p_i x^i$ with $p_i \in \mathbb{R}$ (**zero polynomial** is 0).
- **Degree:** Largest i with $p_i \neq 0$ (degree of zero poly is -1).

Subspaces

Def 4.8 Subspace: Nonempty subset $U \subseteq V$ closed under operations:

1. **Additivity:** $v, w \in U \Rightarrow v + w \in U$
2. **Scaling:** $\lambda \in \mathbb{R}, v \in U \Rightarrow \lambda v \in U$

Lem 4.9 Subspace **must contain zero**: $0 \in U$.

Lem 4.14 U is itself a **vector space** (inherits operations).

Lem 4.11 - 4.13 For $A \in \mathbb{R}^{m \times n}$, these are **Matrix Subspaces**:
Column- $(C(A) \subseteq \mathbb{R}^m)$, Row- $(R(A) \subseteq \mathbb{R}^n)$, Nullspace $(N(A) \subseteq \mathbb{R}^n)$

- **Geometric Examples:** Lines or planes **through the origin** in $\mathbb{R}^3$.
- **Polynomials:** Polys up to degree k ; Polys with no constant term.
- **Matrices:** Symmetric matrices; Matrices with trace 0.

Linearity

Def 4.15 / 4.16 Linear Comb.: $\sum_{j=1}^n \lambda_j v_j \in V$ of **finite** set of vectors.

Def 4.17 Linear **dependent** if some $v \in G$ is a lin. comb. of others ($G \setminus \{v\}$). Otherwise **linear independent**.

- **Span(G):** Set of all (finite) linear combinations of G .

Basis

Def 4.18 Basis: Subset $B \subseteq V$ is a basis if:
 B is **linearly independent** and $\text{Span}(B) = V$.

Thr 4.29 Unique Representation: If B is a basis, every $v \in V$ has **unique** scalars λ_i : $v = \sum \lambda_i v_i$ (defines coordinate system).

Def 4.21 V is **finitely generated** if finite $G \subseteq V$ has $\text{Span}(G) = V$.

Thr 4.22 Every finitely generated space V has a finite basis $B \subseteq G$.

Lem 4.23 Steinitz Exchange Lem.: For finitely generated space V . If finite independent set F and spanning set G ($\text{Span}(G) = V$) then:
1. $|F| \leq |G|$
2. Exist a subset $E \subseteq G$ with $|E| = |G| - |F|$ s.t $\text{Span}(F \cup E) = V$

Obs 4.19 Any m **independent vectors** from basis of $\mathbb{R}^m$.

Lem 4.20 The **independent columns** of A form basis of $C(A)$.

- **Finding Basis:** Remove dependent vectors from spanning set.
- **Zero Space:** Basis of $\{0\}$ is empty set $\emptyset$.

Dimension & Isomorphism

Thr 4.24 All bases of finitely generated space have **same size**.

Def 4.25 Dimension: $\dim(V) = |B|$ (size of basis).

Lem 4.30 Any set G with $|G| < \dim(V)$ **cannot span** V .

- $\dim(\mathbb{R}^m) = m$ and $\dim(\{0\}) = 0$.
- **Sum Rule:** $\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V)$.

Def 4.26 Linear Transf.: Function $T : V \rightarrow W$ preserving structure:

$$T(\lambda_1 x_1 + \lambda_2 x_2) = \lambda_1 T(x_1) + \lambda_2 T(x_2)$$

Def 4.27 Isomorphism: Bijective linear transformation $T : V \rightarrow W$.
• V and W are **isomorphic** if such T exists.

Lem 4.28 Basis B of $V \Leftrightarrow T(B)$ **basis** of W . $\Rightarrow \dim(V) = \dim(W)$

- All **spaces of same dimension** are isomorphic (use basis of both).

4b Computing Matrix Subspaces

- **Fundamental Subspaces** defined as Column-, Row-, and Nullspace.
- Use **Gauss-Jordan** to compute RREF R .

$$A = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{pmatrix} \xrightarrow{\text{RREF}} R = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Thr 4.31 Pivot columns of **original matrix** A form basis for $C(A)$.
 $\Rightarrow \dim(C(A)) = \text{rank}(A) = r$.

- **Example** (using R above): Basis is: $\{(1, 2, 3)^T, (0, 1, 2)^T\}$.

Thr 4.32 First r rows (non-zero rows) of R form a basis for $R(A)$.
 $\Rightarrow \dim(R(A)) = \text{rank}(A) = r$.

- **Example** (using R above): Basis is $\{(1, 2, 0, 3)^T, (0, 0, 1, -2)^T\}$.

Thr 4.33 / 4.34 For any A : $\text{rank}(A) = \text{rank}(A^T)$ and $r \leq \min(m, n)$

Nullspace N(A)

Lem 4.35 Nullspace Isomorphism: $N(A)$ is isomorphic to $\mathbb{R}^{n-r}$.

1. Map $T : N(R) \rightarrow \mathbb{R}^{n-r}$ sends x to vector of its **free variables**.
2. $\dim(N(A)) = n - r$ (number of free variables).

- **Invariant:** $N(A) = N(R)$.
- **Free Variables:** Correspond to non-pivot columns.

Thr 4.36 A basis for $N(A)$ consists of $n - r$ **special solutions**:

- For each free variable k , set $x_k = 1$ and other free vars to 0.
- Solve for dependent variables.

- **Example** (using R above): $(x_1, x_3$ dependent; x_2, x_4 free).
 - Equations: $x_1 = -2x_2 - 3x_4$ and $x_3 = 2x_4$.
 - **Special Sol 1** ($x_2 = 1, x_4 = 0$): $v_1 = (-2, 1, 0, 0)^T$.
 - **Special Sol 2** ($x_2 = 0, x_4 = 1$): $v_2 = (-3, 0, 2, 1)^T$.

4c The General Solution of Ax = b

Def 4.37 The **solution space** is $\text{Sol}(A, b) := \{x \in \mathbb{R}^n : Ax = b\}$.

Thr 4.38 / 4.39

- **Affine:** If s is sol. ($As = b$), then $\text{Sol}(A, b) = \{s + x : x \in N(A)\}$.
- **Dimension:** If solvable, $\dim(\text{Sol}(A, b)) = \dim(N(A)) = n - r$.

- **Never a subspace** if $b \neq 0$ (does not contain zero vector).

Existence & Classification

• **Existence:** Solvable $\Leftrightarrow b \in C(A)$.

Thr 4.40 If A has **full row rank** ($r = m$), then **always solvable**.

Thr 4.41 - 4.44 For **typical** A and b :

- **Rank** $r < m$: Typically unsolvable.
- **Underdetermined** ($m < n$, **Wide**): Typically solvable.
- **Overdetermined** ($m > n$, **Tall**): Typically unsolvable.

5a Orthogonality

Def 5.1.1 $v, w \in \mathbb{R}^n$ are **orthogonal** ($\perp$) $\Leftrightarrow v^T w = 0$.
• V, W **orthogonal** $\Leftrightarrow \forall v \in V, w \in W : v \perp w$.

Lem 5.1.2 Spaces $V \perp W$ (**bases** $\{v_i\}, \{w_j\}$) $\Leftrightarrow v_i \perp w_j$ for all i, j .

- Lem 5.1.3 / 5.1.4** If $V \perp W$:
- Union of bases is **linearly independent**
 - $V \cap W = \{0\}$ (Intersection is zero vector).
 - $\dim(V + W) = \dim(V) + \dim(W) \leq n$.

The Orthogonal Complement

Def 5.1.5 Orthogonal Complement $V^\perp$ is set of all vectors orth. to every vector in V : $V^\perp = \{x \in \mathbb{R}^n \mid v^T x = 0 \quad \forall v \in V\}$

- $V^\perp$ is a subspace.

Lem 5.1.8 $(V^\perp)^\perp = V$.

Thr 5.1.6 / 5.1.9 Compute Complement: For matrix $A \in \mathbb{R}^{m \times n}$:
 $N(A) = C(A^T)^\perp = R(A)^\perp$ and $R(A) = C(A^T) = N(A)^\perp$.

Thr 5.1.7 Decomposition: For **orthogonal** subspaces $V, W \subset \mathbb{R}^n$:
 $W = V^\perp \Leftrightarrow \dim(V) + \dim(W) = n \Leftrightarrow$
Every $u \in \mathbb{R}^n$ has **unique** decomp $u = v + w$ ($v \in V, w \in W$).

Lem 5.1.10 $N(A) = N(A^T A)$ and $C(A^T) = C(A^T A)$.

5b Projections

Def 5.2.1 Projection of b onto subspace S is the closest point $p \in S$:

$$p = \text{proj}_S(b) = \arg \min_{p \in S} \|b - p\|$$

- **Error vector** $e = b - p$ is orthogonal to S ($e \in S^\perp$).
- **Goal:** Best approx. solution for unsolvable $Ax = b$.

Lem 5.2.2 1D Projection of b onto line $\text{Span}(a)$: $p = \frac{a a^T}{a^T a} b$

- Orthogonality condition from error $A^T(b - p) = 0$ leads to:

Lem 5.2.3 General Projection is $p = A\hat{x}$ such that:

$$A^T A \hat{x} = A^T b \quad (\text{Normal Equation})$$

Lem 5.2.4 $A^T A$ is **invertible** $\Leftrightarrow$ columns of A are **linearly indep.**.

Thr 5.2.5 Projection $p = Pb$ uses matrix: $P = A(A^T A)^{-1} A^T$

Obs 5.2.6 Idempotent: $P^2 = P$ (Projecting twice = once).

- **Result:** If $v \in S \Rightarrow Pv = v$; If $v \in S^\perp \Rightarrow Pv = 0$.
- **Complement:** $I - P$ is the projection matrix onto $S^\perp$.

6a Least Squares Approximation

- **Problem:** Find $\hat{x} = \arg \min_x \|Ax - b\|^2$. (☹️ irrelevant as monotonic)

Fact 6.1.1 Minimizer is **sol. of normal equation**. When A has **indep. columns** the **unique** minimizer $\hat{x}$ is: $\hat{x} = (A^T A)^{-1} A^T b$

Applications: Data Fitting

- **Goal:** Fit line $b \approx \alpha_0 + \alpha_1 t$ to points (t_k, b_k) .
- **Matrix Setup:** Unknowns $\hat{x} = (\alpha_0, \alpha_1)^T$, Known $A = [1 \mid t]$ and b .

Lem 6.1.2 Cols of $A = [1 \mid t]$ are **dependent** $\Leftrightarrow t_i = t_j$ for all $i \neq j$.

- **Trick:** Shift to center $t_k^{\text{new}} = t_k - \text{mean}(t)$ (makes $A^T A$ diagnoal)

Rmk 6.1.3 If cols pairwise orthogonal ($\sum t_k = 0$), $A^T A$ is diagonal. Coefficients calculation is independent:

$$\begin{pmatrix} \alpha'_0 \\ \alpha'_1 \end{pmatrix} = \begin{pmatrix} \frac{1}{m} \sum b_k \\ (\sum t_k b_k) / (\sum t_k^2) \end{pmatrix} \text{ and } \alpha_0 = \alpha'_0 - \alpha_1 \text{mean}(t)$$

- **Polynomial Fitting:** Fit curves (e.g. $b \approx \alpha_0 + \alpha_1 t + \alpha_2 t^2$).
- Problem is **linear** in coefficients: $A = [1 \mid t \mid t^2 \mid \dots]$

6b Geometry of Solutions

Lem 6.2.1 Injectivity $C(A^T)$ For $x, y \in C(A^T)$: $Ax = Ay \Leftrightarrow x = y$.

Thr 6.2.2 / 6.2.3 If system $Ax = b$ is **solvable**, there exists a **unique** $x_1 \in C(A^T)$ s.t. $Ax_1 = b$. And $S = x_1 + N(A)$ is the general sol.

- **Min-Norm:** x_1 has the **shortest length** (closest to origin) of all sol.

Thr 6.2.4 Unsolvability Certificate: Exactly one set is non-empty.

1. Solvable: $P = \{x \in \mathbb{R}^n : Ax = b\} \neq \emptyset$.
2. Unsolvability: $D = \{z \in \mathbb{R}^m : A^T z = 0, b^T z = 1\} \neq \emptyset$.

- **Geometric Intuition:** $b \notin C(A) \Rightarrow z \perp C(A)$ and $z \not\perp b$ ($z \in D$)

6c Orthonormal Bases

Def 6.3.1 Orthonormal Vectors $q_1, \dots, q_n$ have length 1 and are pairwise orthogonal: $q_i^T q_j = \delta_{ij}$

Def 6.3.3 Orth. Matrix: Square $Q \in \mathbb{R}^{n \times n}$ with orthonormal col. $\Leftrightarrow Q^T Q = I (= QQ^T)$. Inverse is $Q^{-1} = Q^T$.

Prop 6.3.6 Preserves norm and angle:

$$\|Qx\| = \|x\| \text{ and } (Qx)^T (Qy) = x^T y$$

- **Examples:** Rotation matrices $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

- Prop 6.3.7 / 6.3.12** If cols of Q are orthonormal basis for subspace S :
- **Projection Matrix** onto S : $P = QQ^T$.
 - **Least Squares** solution to $Ax = b$: $R\hat{x} = Q^T b$.

Gram-Schmidt & QR Decomposition

Alg 6.3.8 / 6.3.9 Gram-Schmidt: Convert independent $a_1, \dots, a_n$ into orthonormal $q_1, \dots, q_n$.

1. **Orthogonalize:** $q'_k = a_k - \sum_{i=1}^{k-1} (a_k^T q_i) q_i$ (rem. proj. onto prev. q_i)
2. **Normalize:** $q_k = \frac{q'_k}{\|q'_k\|}$.

Result: $\text{Span}(a_1 \dots a_k) = \text{Span}(q_1 \dots q_k)$ at every step.

Def 6.3.10 / 6.3.11 QR Decomp. For $A \in \mathbb{R}^{m \times n}$ **full rank**, $A = QR$:
• $Q \in \mathbb{R}^{m \times n}$: Orthonormal columns (from Gram-Schmidt).
• $R \in \mathbb{R}^{n \times n}$: **Upper triangular** and **invertible** ($R = Q^T A$).

6d Pseudoinverse $A^\dagger$

Case 1: Full Column Rank ($\text{rank}(A) = n$)

Overdetermined systems ($m \geq n$). $A^\dagger b$ is **Least Squares** solution.

Def 6.4.1 For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = n$: $A^\dagger = (A^T A)^{-1} A^T$

Lem 6.4.2 Left Inverse: If $\text{rank}(A) = n$, then $A^\dagger A = I$.

Case 2: Full Row Rank ($\text{rank}(A) = m$)

Underdetermined systems ($m \leq n$).

Def 6.4.3 For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$: $A^\dagger = A^T (A A^T)^{-1}$

Lem 6.4.4 Right Inverse: If $\text{rank}(A) = m$, then $A A^\dagger = I$.

Lem 6.4.5 / 6.4.6 Min Norm Sol.: For $b \in C(A)$, the **unique** sol. to $\min \|x\|^2$ s.t. $Ax = b$ is $\hat{x} = A^\dagger b$. Strictly in the **Row Space** $C(A^T)$.

Case 3: General Rank ($\text{rank}(A) = r$)

Def 6.4.7 CR Decomposition: Given $A = CR$ (C full col rank, R full row rank): $A^\dagger = R^\dagger C^\dagger$ (order reversed)

Lem 6.4.9 General Factorization: For any $A = ST$ (S full col rank, T full row rank): $A^\dagger = T^\dagger S^\dagger$

Lem 6.4.8 Optimization Meaning: The unique solution to $\min_x \|x\|^2$ s.t. $A^T A x = A^T b$ is $\hat{x} = A^\dagger b$.
Minimizes $\|x\|^2$ among all Least Squares solutions.

Properties

- Always minimizes error $\|Ax - b\|^2$ and then $\|x\|^2$.
- **Left Action** ($b \in \mathbb{R}^m$): $A^\dagger b$ solves $Ax \approx b$.
- **Right Action** ($v^T \in (\mathbb{R}^n)^*$): $v^T A^\dagger$ solves $x^T A \approx v^T$.

Thr 6.4.10 $AA^\dagger A = A$ and $A^\dagger A A^\dagger = A^\dagger$ and $(A^T)^\dagger = (A^\dagger)^T$
• $AA^\dagger$ is symmetric and projects onto **Column Space** $C(A)$
• $A^\dagger A$ is symmetric and projects onto **Row Space** $C(A^T)$

7 Determinant

- **Geometry:** Factor by which linear transf. inflates space. (**Sign** = flip)

Def (lecture) Axioms: Unique function satisfying:

1. **Normalization:** $\det(I) = 1$.
2. **Alternating:** Two identical cols. $\Rightarrow \det(A) = 0$ (implies lin. dep.).
3. **Multilinear:** Linear in each col./row individually (see Prop 7.3.7).

Def 7.1.1 2x2 Determinant: For $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$, $\det(A) = ad - bc$.

Def 7.2.3 Leibniz Formula: For $A \in \mathbb{R}^{n \times n}$, sum over all permuts. Π_n

$$\det(A) = \sum_{\sigma \in \Pi_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i, \sigma(i)}$$

- **Sarrus Rule:** $n!$ terms derived from general formula.

Thr 7.2.5 Symmetry: $\det(A^T) = \det(A)$. (equal over rows).

Prop 7.3.7 Linearity: Det. is linear in each row (or col):

$$\det(\dots, \alpha a + \beta b, \dots) = \alpha \det(\dots, a, \dots) + \beta \det(\dots, b, \dots)$$

Permutations

- **Permutation** ($\sigma \in \Pi_n$): Bijective mapping $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$
- **Swaps:** Transposition (swapping 2 elements) flips sign.

Def 7.2.1 Inversions: $S_I = \{(i, j) : 1 \leq i < j \leq n \text{ and } \sigma(i) > \sigma(j)\}$

Sign: Parity of **inversions**. $\text{sgn}(\sigma) = 1$ if $|S_I|$ is even else -1 .

Fundamental Properties & Operations

- **Similarity:** $\det(SDS^{-1}) = \det(D)$.

Thr 7.1.2 / 7.1.3 / 7.2.6 Multiplicativity & Invertibility:

- $\det(AB) = \det(A) \det(B)$.
- A is **invertible** $\Leftrightarrow \det(A) \neq 0$.
- If invertible: $\det(A^{-1}) = \frac{1}{\det(A)}$.

Prop 7.3.6 Swaps: If P swaps two rows/cols, $\det(PA) = -\det(A)$.

Prop 7.2.4 Special Matrix Classes

1. **Permutation** (P): $\det(P) = \text{sgn}(\sigma)$ (based on swap count).
2. **Triangular** (T): $\det(T) = \prod_{k=1}^n T_{kk}$ (product of diagonals).
3. **Orthogonal** (Q): $Q^T Q = I \Rightarrow \det(Q) \in \{1, -1\}$.

Cofactors & Applications

Def 7.3.1 Cofactor: $C_{ij} = (-1)^{i+j} \det(\mathcal{A}_{ij})$ ($\mathcal{A}_{ij}$ is w/o row i , col j).

Prop 7.3.2 Laplace Expansion row (or col) i : $\det(A) = \sum_{j=1}^n A_{ij} C_{ij}$

Prop 7.3.3 Inverse (if exists): $A^{-1} = \frac{1}{\det(A)} C^T$ (C is cofactor matrix).

Prop 7.3.5 Cramer's Rule: Solution to $Ax = b$: $x_j = \frac{\det(\mathcal{B}_j)}{\det(A)}$
 $\mathcal{B}_j$ is A with col j replaced by b .

Efficient Calculation

- **Gaussian Elimination:**
 1. Transform $A \rightarrow$ upper triangular U (track swaps k)
 2. $\det(A) = (-1)^k \cdot \prod_{i=1}^n U_{ii}$.
- **Laplace Expansion:** Recursive. Good for **sparse** matrices.

3c Eigenvalues and Eigenvectors

Complex Numbers Foundation

- **Definition:** $\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}$ where $i^2 = -1$.
- **Parts:** $\Re(z) = a, \Im(z) = b$
- **Modulus:** $|z| = \sqrt{a^2 + b^2}$ **Conjugate:** $\overline{a + ib} = a - ib$.
- **Division:** Expand by conjugate $\frac{z}{w} = \frac{z \bar{w}}{|w|^2} = \frac{(ax+by)+i(bx-ay)}{x^2+y^2}$.
- **Polar Form:** $z = re^{i\theta}$ ($r = |z|$, θ is argument).

Rmk 8.1.1 Euler's Formula: $e^{i\theta} = \cos \theta + i \sin \theta$ and $e^{i\pi} = -1$.

Thr 8.1.2 / 8.1.3 Fund. Thr. of Algebra: $\mathbb{C}$ is **algebraically closed**.

Any degree $n \geq 1$ polynomial $P(z)$ has n roots $\lambda_1, \dots, \lambda_n \in \mathbb{C}$:

$$P(z) = \alpha_n (z - \lambda_1) \dots (z - \lambda_n)$$

Algebraic Multiplicity: Number of times λ appears as a root.

Complex Vectors and Matrices

- **Conjugate Transpose** (Hermitian): $A^H = (\bar{A})^T$ ($(A^H)_{ij} = \bar{A}_{ji}$).
- Properties: $(AB)^H = B^H A^H$ and $(A^H)^H = A$.
- **Hermitian Matrix:** $A = A^H$ (Complex analog to Symmetric).
- Diagonal entries must be real ($A_{ii} \in \mathbb{R}$).
- **Unitary Matrix:** U is Unitary if $U^H U = I \Leftrightarrow U^{-1} = U^H$.
- Analog to Orthogonal. Columns form **orthonormal** basis for $\mathbb{C}^n$.
- **Complex Inner Product:** $\langle u, v \rangle = u^H v = \sum_{i=1}^n \bar{u}_i v_i$ (**not** commut.).
- **Complex Norm:** $\|v\| = \sqrt{\langle v, v \rangle} = \sqrt{|v_1|^2 + \dots + |v_n|^2}$.

Eigenvalues and Eigenvectors

Def 8.2.1 For $A \in \mathbb{R}^{n \times n}$, $\lambda \in \mathbb{C}$ is an **eigenvalue** and $v \in \mathbb{C}^n \setminus \{0\}$ is an **eigenvector** if $Av = \lambda v$. (λ, v) is an **eigen-pair**.

- **Real λ :** Scaling. **Complex λ :** Rotation + Scaling.

Lem 8.2.3 Real Eigenvalues

- $\lambda \in \mathbb{R}$ is an eigenvalue $\Leftrightarrow \det(A - \lambda I) = 0$.
- $v \in \mathbb{R}^n \setminus \{0\}$ is eigenvector $\Leftrightarrow v \in N(A - \lambda I)$.

Prop 8.2.4 Characteristic Polynomial: $P(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Thr 8.2.5 Existence: Every $A \in \mathbb{R}^{n \times n}$ has at least one eigenvalue.

Lem 8.2.8 Conjugate Pairs: If (λ, v) is an eigenpair of real matrix A , then $(\bar{\lambda}, \bar{v})$ is also an eigenpair.

Properties and Relations

Def 8.3.4 The **trace** is the sum of the diagonal: $\text{Tr}(A) = \sum_{i=1}^n A_{ii}$.

Lem 8.3.7 Cyclic Trace: $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$.

Lem 8.3.6 Magic Connections: For eigenvalues $\lambda_1, \dots, \lambda_n$ of A :
Determinant: $\det(A) = \prod_{i=1}^n \lambda_i$ and **Trace:** $\text{Tr}(A) = \sum_{i=1}^n \lambda_i$

Lem 8.3.5 Transpose: A and A^T have the **same eigenvalues**.

- Prop 8.3.1** If (λ, v) is an eigenpair of A :
- **Powers:** (λ^k, v) is eigenpair of A^k ($k \geq 1$).
 - **Inverse:** $(\frac{1}{\lambda}, v)$ is eigenpair of A^{-1} (if invertible).

- If μ is eigenvalue of A^k , then some $\sqrt[k]{\mu} \in \mathbb{C}$ is eigenvalue of A .

Special Matrix Classes

Prop 8.2.7 Orthogonal: If Q is orthogonal, all eigvals $\lambda \in \mathbb{C}$: $|\lambda| = 1$.

- **Rotation:** $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (90° rot) has $\lambda = \pm i$ (pure imaginary).
- For $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$, eigenvalues are union of eigenvalues of A and D .

Linear Independence

Lem 8.3.2 Eigenvectors to distinct eigenvalues are **lin. indep.**.

Thr 8.3.3 If A has n distinct real eigval. $\Rightarrow$ Eigvectors basis for $\mathbb{R}^n$.

Applications

Fibonacci Numbers

Fibonacci Recurrence: $F_{n+1} = F_n + F_{n-1}$ as system $g_n = M g_{n-1}$ with $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $g_0 = (1, 0)^T$.

1. **Find Eigenpairs:** φ_i, v_i (in this case $\varphi_0 = \frac{1+\sqrt{5}}{2}$ the Golden Ratio).
2. **Decompose** start g_0 into eigenbasis: $g_0 = c_1 v_1 + c_2 v_2$.
3. **Apply powers:** $g_n = M^n g_0 = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$.
4. Extract F_n from g_n for **closed form**.

Stability & Power Method

- For large n , $A^n x$ is **dominated** by largest $|\lambda_{\max}|$.
- All $|\lambda| < 1 \Rightarrow$ **Decay** (converges to 0).
- Any $|\lambda| > 1 \Rightarrow$ **Growth** (diverges).

9a Diagonalization

Def 9.1.2 Diagonalizable Matrix: $A \in \mathbb{R}^{n \times n}$ is diagonalizable $\Leftrightarrow \exists$ invertible V s.t. $V^{-1}AV = \Lambda$ (diagonal).

Thr 9.1.1 Eigendecomposition: A has eigenvectors $v_1, \dots, v_n$ forming a basis of $\mathbb{R}^n \Leftrightarrow A = V\Lambda V^{-1}$.

- $V = (v_1 \dots v_n)$ (columns are eigenvectors).
- Λ : Diagonal matrix with $\Lambda_{ii} = \lambda_i$ (associated eigenvalues).

- Represents linear transformation as **scaling** in the **eigenbasis**.

Conditions for Diagonalizability

Def 9.1.10 For eigvalue λ of A , the **geometric multiplicity** is the dimension of the eigenspace: $\dim(N(A - \lambda I))$.

- **Algebraic Mult.:** Mult. of root λ in char. polynomial $\det(A - \lambda I)$.

Def 9.1.3 A has **complete set of real eigvecs.** if they form $\mathbb{R}^n$ basis.

Lem 9.1.11 Criterion: A has a complete set (is diagonalizable) $\Leftrightarrow$ All eigenval are **real** and all **Geom. Multiplicity = Alg. Multiplicity**.

- If A has n distinct real eigval $\Rightarrow$ diagonalizable (eigvecs. lin. indep.).

Matrix Classes & Examples

- **Diagonal Matrices:** eigvalues D_{ii} ; eigvectors. $e_1, \dots, e_n$ form basis

Prop 9.1.6 Projection Mat. P on space U always **diagonalizable**.
Eigvals: 0, 1 **Basis:** Union of basis for U ($\lambda = 1$) and $U^\perp$ ($\lambda = 0$).

- **Nilpotent Matrix** ($A^k = 0$): Only diagonalizable if $A = 0$.

Similarity Transformations

Def 9.1.7 A, B are **similar** if $\exists$ invertible S s.t. $B = S^{-1}AS$.

Prop 9.1.8 If A and B are similar $\Rightarrow$ **same eigvals** (same char. poly)
• A has complete set (diagonalizable) $\Leftrightarrow B$ does.

- **Eigenvectors:** Not identical; transform via S ($v_B = S^{-1}v_A$).

9b Symmetric Matrices

Lem 9.2.8 / 9.2.9 Let $A \in \mathbb{R}^{n \times n}$ be **symmetric** ($A = A^T$).
• **Only real Eigenvalues:** All $\lambda \in \mathbb{R}$. At least one eigvalue exists.

Lem 9.2.4 / 9.2.5 / 9.2.7 For symmetric A :

- Eigvecs to **distinct** eigvals ($\lambda_1 \neq \lambda_2$) are **orthogonal** ($v_1 \perp v_2$).
- **Alg. mult. = Geom. mult.** for all eigvals. (always **complete set**)
- $\text{rank}(A)$ = the number of **non-zero eigenvalues** (with repetitions).

Spectral Theorem & Eigendecomposition

Thr 9.2.1 Spectral Theorem: Any symmetric matrix A has n real eigenvalues and an **orthonormal basis** of $\mathbb{R}^n$ of eigenvectors.

Cor 9.2.2 / 9.2.3 Eigendecomposition: Symmetric A is diagonalizable by an **orthogonal matrix** V ($V^T = V^{-1}$): $A = V\Lambda V^T$
• Λ : Eigvalues on diagonal. V : Columns are orthonormal eigenvectors.

Prop 9.2.6 A is a weighted sum of **rank-1 projections** onto eigenspaces: $A = \sum_{i=1}^n \lambda_i v_i v_i^T$

- Transformation is Rotation (V^T) $\rightarrow$ Scaling (Λ) $\rightarrow$ Rotation back (V).

Positive Definite (PD) & Semidefinite (PSD)

Prop 9.2.10 For sym. A and $x \neq 0$ **Rayleigh Quotient:** $R(x) = \frac{x^T A x}{x^T x}$
 $R(x)$ is **bounded** by eigenvalues of A : $\lambda_{\min} \leq R(x) \leq \lambda_{\max}$

Def 9.2.11 / 9.2.12 For sym. A with eigenvalues $\lambda_1, \dots, \lambda_n$:
• **Positive Semidefinite (PSD):** All $\lambda_i \geq 0 \Leftrightarrow x^T A x \geq 0$ ($\forall x$).
• **Positive Definite (PD):** All $\lambda_i > 0 \Leftrightarrow x^T A x > 0$ ($\forall x \neq 0$).

- **Diagonal Dominance:** If $|A_{ii}| > \sum_{j \neq i} |A_{ij}|$ is often PD/PSD.
- **Sum:** PSD + PSD = PSD; PD + PSD = PD; PD + PD = PD.
- **Proof:** $x^T(A+B)x = x^T A x + x^T B x \geq 0$ or > 0 .
- **Product** (not symmetric): PSD $\times$ PD = $\lambda_i \geq 0$; PD $\times$ PD = $\lambda_i > 0$.
- **Proof** (A PD, $A = RR^T$): AB similar to $C = R^{-1}ABR = R^T B R$.
 $x^T C x = x^T R^T B R x = (R x)^T B (R x) = y^T B y \geq 0$ or > 0
- **Inverse:** PD always invertible (0 not eigval). **PSD** may be singular.

Gram Matrices & Cholesky

Def 9.2.13 Gram Matrix of vecs $v_1, \dots, v_n$ (cols of V) is $G = V^T V$.
• Entries are inner products: $G_{ij} = v_i^T v_j$.

Obs 9.2.14 Abuse of not.: $AA^T = \sum a_i a_i^T$ also called Gram matrix.

Prop 9.2.15 For any $A \in \mathbb{R}^{m \times n}$: $A^T A$ ($n \times n$) and AA^T ($m \times m$) are **symmetric** and **PSD** and share **same non-zero eigenvalues**.

Prop 9.2.16 Cholesky Decomposition: Every symmetric **PSD** is Gram matrix $M = C^T C$ where C is **upper triangular**.

Construction: 1. Diagonalize $M = V\Lambda V^T$
2. quare root eigenvalues: $\Lambda^{\frac{1}{2}}$. 3. Form $C = (\Lambda^{\frac{1}{2}} V) (\Lambda^{\frac{1}{2}} V)^T$
4. Apply QR decomp to $(\Lambda^{\frac{1}{2}} V)^T \Rightarrow M = (QR)^T (QR) = R^T R$

9c Singular Value Decomposition

Def 9.3.1 / 9.3.3 Every matrix $A \in \mathbb{R}^{m \times n}$ has an **SVD:** $A = U\Sigma V^T$
• $U \in \mathbb{R}^{m \times m}$: Orthogonal ($U^T U = I$). Col. u_i **Left Singular Vectors**.
• $V \in \mathbb{R}^{n \times n}$: Orthogonal ($V^T V = I$). Col. v_i **Right Singular Vectors**.
• $\Sigma \in \mathbb{R}^{m \times n}$: Rectangular diagonal matrix. Entries σ_i (**Singular Values**) are non-negative and ordered: $\sigma_1 \geq \dots \geq \sigma_{\min} \geq 0$.

- **Identities:** $AV = U\Sigma$ and $Av_i = \sigma_i u_i$

Obs 9.3.2 Compact SVD: Efficient form for rank r : $A = U_r \Sigma_r V_r^T$
• **rank(A)** is the number of **non-zero** singular values ($\sigma_r > 0$).
• $U_r \in \mathbb{R}^{m \times r}$ (first r cols), $V_r \in \mathbb{R}^{n \times r}$ (first r cols).
• $\Sigma_r \in \mathbb{R}^{r \times r}$ (diagonal, strictly positive).

- **Storage:** Compact form stores $r(m+n+1)$ entries (vs mn for full).
- **Compact $\rightarrow$ Full:** Σ : Pad with zeros. U / V : Extend to orthonormal basis (e.g. Gram-Schmidt).

Construction & Relations

- **From any A (General):** 1. Eigendecomp $A^T A \rightarrow$ sorted λ_i, v_i .
2. **Right site:** $V = [v_1, \dots, v_n]$. **Singular Values:** $\sigma_i = \sqrt{\lambda_i}$.
3. **Left side** (only $i \leq r$): $U_r = AV_r \Sigma_r^{-1}$ or $u_i = \frac{1}{\sigma_i} A v_i$.
4. **Extend:** If $m > r$, fill rest of U via Gram-Schmidt (spans $N(A^T)$).
- **From Spectral Decomposition** (Symmetric $A = Q\Lambda Q^T$):
1. Σ : $\sigma_i = |\lambda_i|$ (absolute values). 2. V : $v_i = q_i$ (orig. eigenvectors).
3. U : $u_i = \text{sgn}(\lambda_i) v_i$ (flip sign if $\lambda_i < 0$). 4. **Sort** cols and diag by σ_i .
- **Rel.:** V : Eigvecs of $A^T A$ / U : Eigvecs of AA^T / Σ : $\sigma_i = \sqrt{\lambda_i}$ of either.

Subspaces & Inverses

- **Column Space** $C(A)$: Span of first r cols of U ($u_1 \dots u_r$).
- **Left Nullspace** $N(A^T)$: Span of last $m-r$ cols of U .
- **Row Space** $C(A^T)$: Span of first r cols of V ($v_1 \dots v_r$).
- **Nullspace** $N(A)$: Span of last $n-r$ cols of V .
- **Pseudoinverse:** $A^\dagger = V\Sigma^\dagger U^T$. $\Sigma^\dagger$: Transpose Σ , invert entries ($\frac{1}{\sigma_i}$).

Applications

Lem 9.3.4 Rank- r Decomposition: Any matrix of rank r is the sum of r outer products: $A = \sum_{k=1}^r \sigma_k u_k v_k^T$

- **Low-Rank Approximation:** Truncate sum at $k < r$ (keep largest σ)

11 Skew-Symmetric Matrices

- **Def:** Square matrix $A \in \mathbb{R}^{m \times m}$ is **skew-symmetric** $\Leftrightarrow A^T = -A$.
- **Entry-wise:** $a_{ij} = -a_{ji}$. (diagonal zero $a_{ii} = 0 \forall i \in [m]$)

Vector Space Properties

- The set S_m of skew-symmetric matrices is a **subspace** of $\mathbb{R}^{m \times m}$.
- **Dimension:** $\dim(S_m) = \frac{m(m-1)}{2}$ (**Basis** by entries **above diagonal**)
- **Determinant & Rank**
 - **Odd Dimension:** If n is **odd**, A is **singular** ($\det(A) = 0$).
 - **Proof:** $\det(A) = \det(-A^T) = (-1)^n \det(A^T) = (-1)^n \det(A)$.
If n odd: $\det(A) = -\det(A) \Rightarrow \det(A) = 0$.
 - **Even Dimension:** If n is **even**, A can be invertible ($\det(A) \neq 0$).
 - **Example:** $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ has $\det(A) = 1$. Build recursive blocks.
- **Rank Parity:** Rank of skew-symmetric matrix is **always even**.
- **Proof:** Non-zero eigenvalues come in conjugate pairs $\lambda = \pm ik$.

Spectral Properties & Algebra

- **Quadratic Form:** Vanishes for all real vectors ($x^T A x = 0$).
- **Proof:** $x^T A x = (x^T A x)^T = x^T A^T x = -x^T A x$.
- $-A^2$ is **symmetric** and **PSD** $\Rightarrow$ eigenvalues of A^2 are ≤ 0 .
- **Proof Sym:** $(-A^2)^T = -(A^T A^T) = -(-A)(-A) = -A^2$.
- **Proof PSD:** $x^T (-A^2) x = x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 \geq 0$.
- **Eigenvalues of A:** Purely **imaginary** or zero ($\lambda \in i\mathbb{R}$).
- **Proof:** $Ax = \lambda x \Rightarrow A^2 x = \lambda^2 x$. Since $\lambda^2 \leq 0 \Rightarrow \lambda \in i\mathbb{R}$.

12 Useful Formulas

Trigonometry

Deg	360° / 0°	30°	45°	60°	90°
Rad	$2\pi / 0$	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
sin	0	1/2	$\sqrt{2}/2$	$\sqrt{3}/2$	1
cos	1	$\sqrt{3}/2$	$\sqrt{2}/2$	1/2	0
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

$$\sin(\alpha) = \frac{\text{opp}}{\text{hyp}}, \quad \cos(\alpha) = \frac{\text{adj}}{\text{hyp}}, \quad \tan(\alpha) = \frac{\text{opp}}{\text{adj}} = \frac{\sin(\alpha)}{\cos(\alpha)}$$

- **Pythagoras:** $\sin^2(x) + \cos^2(x) = 1$
- **Symmetry:** $\sin(-x) = -\sin(x)$, $\cos(-x) = \cos(x)$
- **Phase:** $\sin(x + \frac{\pi}{2}) = \cos(x)$
- **Addition:** $\sin(x \pm y) = \sin(x) \cos(y) \pm \cos(x) \sin(y)$
 $\cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y)$

Quadratic Formula

- For $ax^2 + bx + c = 0$ ($a \neq 0$): $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Discriminant** $D = b^2 - 4ac$:
 - $D > 0$: 2 real solutions.
 - $D = 0$: 1 real solution (multiplicity 2).
 - $D < 0$: 2 complex solutions (conjugate pairs).

Binomials

- $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$
- **Theorem:** $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$
- **Coefficient:** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

Sums

- **Gaussian Sum:** $\sum_{i=1}^n i = (n(n+1))/2$.
- **Sum of Cubes:** $\sum_{i=1}^n i^3 = (n^2(n+1)^2)/4$.

12 Multiple Choice

General & Linear Maps

- ✓ $\text{Tr}(A)$ is linear; $\text{Tr}(AB) = \text{Tr}(BA)$.
- ✗ $\det(A), \|A\|$ are linear.
- ✓ A^k is invertible $\Leftrightarrow A$ is invertible (for $k \in \mathbb{Z}$).
- ✓ $A^2 = A$ (Idempotent) and A invertible $\Rightarrow A = I$.
- ✓ $(AB)^{-1} = B^{-1}A^{-1}$ (if A, B invertible).
- ✓ $(A^T)^{-1} = (A^{-1})^T$.
- ✗ $ABA^{-1} = B$ (generally false, unless $AB = BA$).
- ✗ x, y, z dependent $\Rightarrow x \in \text{span}(y, z)$ (could be $y \in \text{span}(z)$).
- ✓ $Ax = 0$ has non-trivial sol $x \neq 0 \Rightarrow A$ singular.
- ✓ $A \in \mathbb{R}^{m \times n}$ has rank(A) = $m \Rightarrow Ax = b$ solvable $\forall b$ (Surjective).

Norms & Dot Products

- ✓ $x^T (A^T y) = (Ax)^T y$ (Adjoint property).
- ✗ $f(x) = \|Ax\|$ is a norm (True **only** if $N(A) = \{0\}$).
- ✗ $\|v\| \leq \|v+u\|$ (Reverse Triangle is $\|v\| - \|u\| \leq \|v-u\|$).
- **Orthogonal Matrix** Q ($Q^T = Q^{-1}$):
 - ✓ Preserves dot product ($(Qx)^T (Qy) = x^T y$).
 - ✓ Isometry ($\|Qx\|_2 = \|x\|_2$).
 - ✓ Q^T, Q^{-1}, AB (if A, B orth) are orthogonal.
- ✗ $A+B$ is orthogonal (almost always false).

Vector Spaces

- ✓ $S \subset V$ and subspace $W \subset V$: $S \subset W \Rightarrow \text{span}(S) \subset W$.
- ✓ Finite dim Euclidean space ($\mathbb{R}^n$) always has orthonormal basis.
- ✓ Any Orthogonal set can be extended to orthonormal basis.
- **Subspaces of $\mathbb{R}^{n \times n}$:**
 - ✓ Symmetric $\{A \mid A^T = A\}$ and Skew-Sym $\{A \mid A^T = -A\}$.
 - ✓ Commuting $\{A \mid AB = BA\}$.
 - ✗ **Invertible** ($\text{GL}(n)$) or **Singular matrices** (not closed under $+$).
- ✓ $\dim(\text{Skew}_{n \times n}) = \frac{n(n-1)}{2}$ (For $n = 3$, $\dim=3$).

Linear Mappings $F : V \rightarrow W$

- ✓ F inj $\Rightarrow F(\text{basis})$ is lin. indep.
- ✓ F surj $\Rightarrow F(\text{basis})$ is generating set.
- ✓ F iso $\Leftrightarrow F(\text{basis})$ is a basis.
- **Orthogonal Map:**
 - ✓ Isomorphism onto its Image.
 - ✓ $F^{-1} : C(F) \rightarrow V$ is orthogonal.
- ✗ $v^T F(v) = 0 \forall v \Rightarrow F = 0$ (False for real matrices, e.g., Rot. 90°).

Determinant

- ✓ $\det(A^k) = \det(A)^k$ and $\det(A^T) = \det(A)$.
- ✓ $\det(PAP) = \det(A)$ if P is Permutation ($P^T P = I$).
- ✓ $\det(PBP^{-1}) = \det(B)$ (Similarity invariant).
- ✗ $\det(cA) = c \det(A)$ (True is $\det(cA) = c^n \det(A)$).
- ✗ $\det(A+B) = \det(A) + \det(B)$.
- ✗ $\det(QA) = \det(A)$ with Q orth. (True is $|\det(QA)| = |\det(A)|$).
- ✓ A singular $\Rightarrow AB$ singular
 - For $AB = -BA$ ($n \times n$):
 - ✓ $\det(AB) = \det(-BA) = (-1)^n \det(BA)$.
 - ✓ If n is odd, $\det(AB) = 0$ (either A or B singular).

Eigenvalues & Diagonalization

- ✓ v is eigenvector $\Rightarrow cv$ is eigenvector ($\forall c \neq 0$).
- ✓ Real Matrix A has eigenvalue $\Rightarrow \infty$ eigenvectors.
- ✓ Distinct $\lambda_i \Rightarrow$ eigenvectors lin. indep $\Rightarrow$ Diagonalizable.
- ✗ **Diagonalizable $\Rightarrow$ distinct λ_i** (e.g., I).
- ✓ A similar $B \Rightarrow \lambda_A = \lambda_B$ (Same char. poly).
- ✗ A similar $B \Rightarrow v_A = v_B$ (Eigenvectors change).
- ✗ v_1, v_2 eigenvectors $\Rightarrow v_1 + v_2$ eigenvectors (True only if $\lambda_1 = \lambda_2$).
- ✓ Rotation in $\mathbb{R}^3$ always has $\lambda = 1$ (Axis).

QR Decompositions

- ✓ A cols lin. indep $\Rightarrow A = QR$ unique (with $\text{diag}(R) > 0$).
- ✓ A cols lin. indep $\Rightarrow \text{rank}(R) = n$ (Full rank).
- ✓ $A = Q_1 R_1 = Q_2 R_2 \Rightarrow Q_1^T Q_2$ is diagonal matrix.

Systems & Subspaces

- ✓ **Dimension Formula:** $\dim(N(A)) + \dim(C(A)) = n$.
- ✓ $Ax = 0$ only trivial sol $\Rightarrow \dim(N(A)) = 0, \text{rank}(A) = n$.
- ✓ $C(A) = C(AA^T)$ and $N(A) = N(A^T A)$.
- ✓ $C(A) = C(2A)$ and $N(A) = N(2A)$.
- ✗ $C(A) = C(A^2)$ (generally false).
- ✗ $N(A) = N(A^T)$ (generally false).

Images

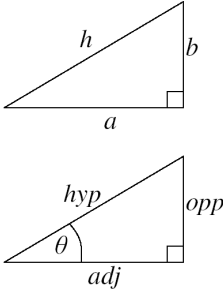
Matrix Multiplication

matrix multiplication	covector / vector / ope	
Example	Example	
$\underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{2 \times 1} = \underbrace{\begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix}}_{3 \times 1}$	$\underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\substack{\in \mathbb{R}^2 \\ \in \mathbb{R}^3}} = \underbrace{\begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix}}_{\in \mathbb{R}^3}$	$\underbrace{Ax}_{\in \mathbb{R}^m}$
$\underbrace{\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}}_{1 \times 2} \underbrace{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}_{2 \times 1} = \underbrace{[11]}_{1 \times 1}$	$\underbrace{\begin{pmatrix} 1 & 2 \end{pmatrix}}_{\in (\mathbb{R}^2)^*} \underbrace{\begin{pmatrix} 3 \\ 4 \end{pmatrix}}_{\in \mathbb{R}^2} = \underbrace{11}_{\in \mathbb{R}}$	$\underbrace{v^T w}_{\in \mathbb{R}}$
$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 4 & 4 & 5 \end{bmatrix}}_{1 \times 3} \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} = \underbrace{\begin{bmatrix} 6 & 9 \end{bmatrix}}_{1 \times 2}$	$\underbrace{\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}}_{\in (\mathbb{R}^3)^*} \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} = \underbrace{\begin{bmatrix} 6 & 9 \end{bmatrix}}_{\substack{\in (\mathbb{R}^2)^* \\ \in (\mathbb{R}^n)^*}}$	$\underbrace{y^T A}_{\in (\mathbb{R}^n)^*}$
$\underbrace{\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}}_{3 \times 1} \underbrace{\begin{bmatrix} 1 & 2 \end{bmatrix}}_{1 \times 2} = \underbrace{\begin{bmatrix} 3 & 6 \\ 4 & 8 \\ 5 & 10 \end{bmatrix}}_{3 \times 2}$	$\underbrace{\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}}_{\substack{\in \mathbb{R}^3 \\ \in (\mathbb{R}^2)^*}} \underbrace{\begin{pmatrix} 1 & 2 \end{pmatrix}}_{\substack{\in (\mathbb{R}^2)^* \\ \in (\mathbb{R}^2)^*}} = \underbrace{\begin{bmatrix} 3 & 6 \\ 4 & 8 \\ 5 & 10 \end{bmatrix}}_{3 \times 2}$	$\underbrace{vw^T}_{m \times n}$
$\underbrace{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}_{2 \times 1} \underbrace{\begin{bmatrix} 2 \end{bmatrix}}_{1 \times 1} = \underbrace{\begin{bmatrix} 6 \\ 8 \end{bmatrix}}_{2 \times 1}$	$\underbrace{\begin{pmatrix} 3 \\ 4 \end{pmatrix}}_{\in \mathbb{R}^2} \underbrace{2}_{\in \mathbb{R}} = \underbrace{\begin{pmatrix} 6 \\ 8 \end{pmatrix}}_{\in \mathbb{R}^2}$	$\underbrace{v\lambda}_{\in \mathbb{R}^m}$
$\underbrace{\begin{bmatrix} 2 \end{bmatrix}}_{1 \times 1} \underbrace{\begin{bmatrix} 3 & 4 \end{bmatrix}}_{1 \times 2} = \underbrace{\begin{bmatrix} 6 & 8 \end{bmatrix}}_{1 \times 2}$	$\underbrace{2}_{\in \mathbb{R}} \underbrace{\begin{pmatrix} 3 & 4 \end{pmatrix}}_{\substack{\in (\mathbb{R}^2)^* \\ \in (\mathbb{R}^2)^*}} = \underbrace{\begin{bmatrix} 6 & 8 \end{bmatrix}}_{\in (\mathbb{R}^2)^*}$	λv^T
$\underbrace{\begin{bmatrix} 2 \end{bmatrix}}_{1 \times 1} \underbrace{\begin{bmatrix} 3 \end{bmatrix}}_{1 \times 1} = \underbrace{\begin{bmatrix} 6 \end{bmatrix}}_{1 \times 1}$	$\underbrace{2}_{\in \mathbb{R}} \underbrace{3}_{\in \mathbb{R}} = \underbrace{6}_{\in \mathbb{R}}$	$\lambda \mu$

Subspaces

$C(A)$	$N(A)$	$R(A)$
	$\underbrace{\begin{pmatrix} x_2, x_4 \\ 1 \\ 0 \end{pmatrix}}_{x_2, x_4} \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{x_2, x_4}$	
Bases:	$\underbrace{\begin{pmatrix} x_1, x_3 \\ -2 \\ 0 \end{pmatrix}}_{x_1, x_3} \underbrace{\begin{pmatrix} -3 \\ 2 \end{pmatrix}}_{x_1, x_3}$	$\underbrace{\begin{pmatrix} 1 \\ 2 \\ 0 \\ 3 \end{pmatrix}}_{x_1, x_3} \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}}_{x_2, x_4}$
$\underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}}_C \leftarrow \underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix}}_{A=CR'} \rightarrow \underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{R \text{ in RREF}(1,3), Q} \rightarrow \underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}}_{R' \text{ in RREF}(1,3)}$		
Theorem 4.31	Theorem 4.36	Theorem 4.32

Triangle



Pythagoras's Theorem

$a^2 + b^2 = h^2$

Trigonometric Ratios

$\sin(\theta) = \frac{opp}{hyp}$

$\cos(\theta) = \frac{adj}{hyp}$

$\tan(\theta) = \frac{opp}{adj}$